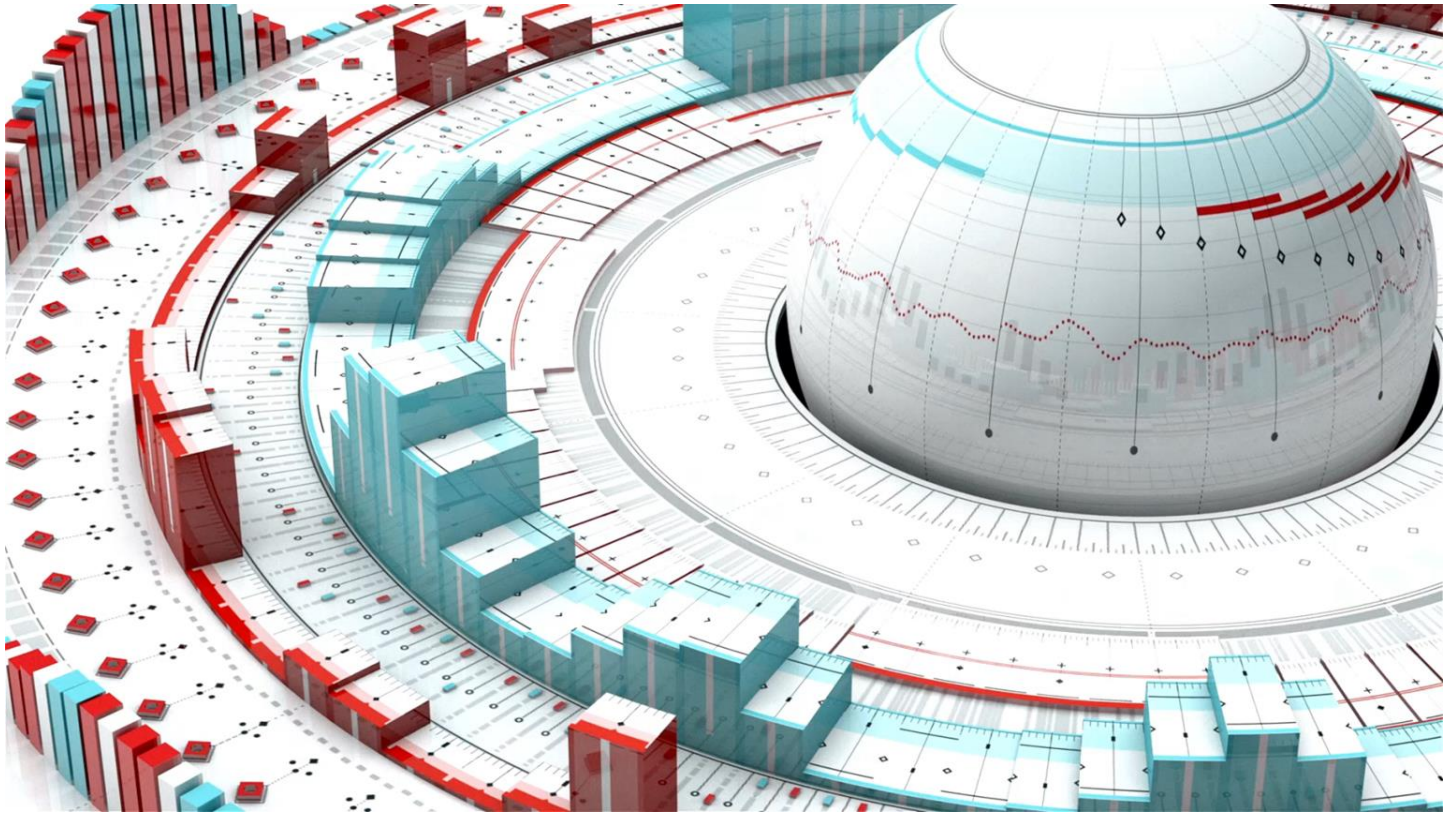
Data Visualization



Data visualization

Data visualization is the graphical representation of information and data. By using visual elements like charts, graphs, and maps, data visualization tools provide an accessible way to see and understand trends, outliers, and patterns in data. Additionally, it provides an excellent way for employees or business owners to present data to non-technical audiences without confusion.

In the world of Big Data, data visualization tools and technologies are essential to analyze massive amounts of information and make data-driven decisions.

Outline

1. What is Data Visualization?
2. Benefits of Good Data Visualization
3. Different Types of Analysis for Data Visualization
4. Univariate Analysis Techniques for Data Visualization
5. Bivariate Analysis Techniques for Data Visualization

Why Data Visualization?

Why use data visualization?

1.Make data easier to understand and remember
2.Discover unknown facts, outliers, and trends
3.Visualize relationships and patterns quickly
4.Ask better questions and make better decisions



What is Data Visualization?

- Data visualization is defined as a **graphical representation** that contains the **information** and the **data**.
- By using visual elements like **charts**, **graphs**, and **maps**, data visualization techniques provide an accessible way to see and **understand trends, outliers, and patterns in data**.
- In modern days we have a lot of data in our hands i.e, in the world of **Big Data**, data visualization tools, and technologies are crucial to analyze massive amounts of information and make data-driven decisions.
- It is used in many areas such as:
  - To model complex events.
  - Visualize phenomenons that cannot be observed directly, such as weather patterns, medical conditions, or mathematical relationships.

Benefits of Good Data Visualization

- It is a powerful technique to explore the data with **presentable** and **interpretable** results.
- In the **data mining process**, it acts as a primary step in the pre-processing portion.
- It supports the **data cleaning process** by finding incorrect data and corrupted or missing values.
- It also helps to **construct and select variables**, which means we have to determine which variable to include and discard in the analysis.
- In the process of **Data Reduction**, it also plays a crucial role while combining the categories.

Types of Data Visualization

Table	Column Chart	Bar Graph	Stacked Bar Graph	Stacked Column Chart	Pie Chart
Histogram	Box Plot	Distributed Graph	Heat Maps	Tree Maps	Mosaic Plot
Scatter plots	Line Graph	Area Chart	Dotplot/strip chart	Waterfall Chart	Bubble Chart

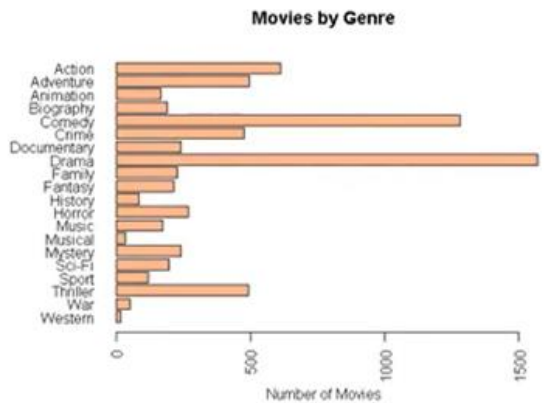
Categorical Univariate Visualization

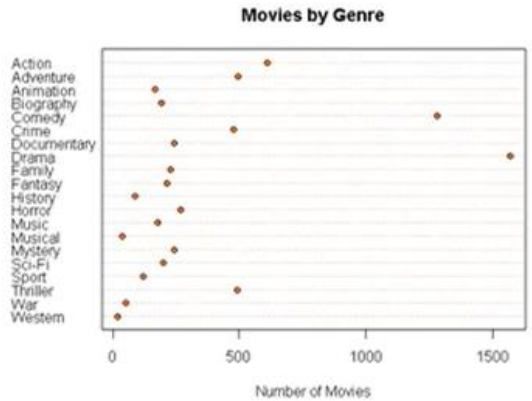
Categorical Univariate Visualization

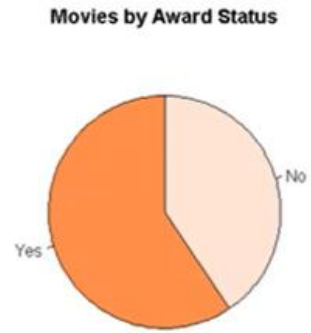
Visualizing One Categorical Variable

Movies by Genre		
Genre	Frequency	Percentage
Action	612	9%
Adventure	496	7%
Animation	168	2%
Comedy	1281	18%
Drama	1570	22%
Horror	269	4%
...	...	...







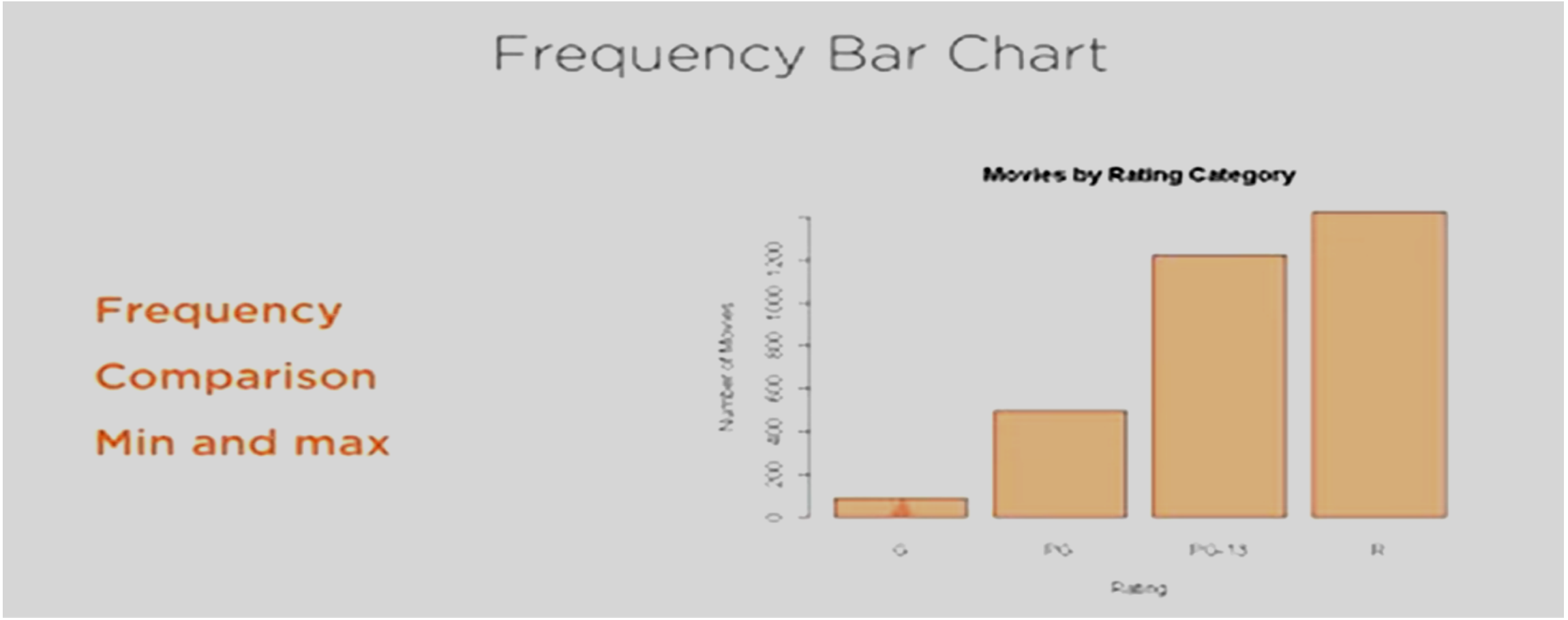


Qualitative Univariate Analysis

Frequency

Proportion

Movies by Genre		
Genre	Frequency	Percentage
Action	612	9%
Adventure	496	7%
Animation	168	2%
Comedy	1281	18%
Drama	1570	22%
Horror	269	4%
...	...	...



Bar Chart:

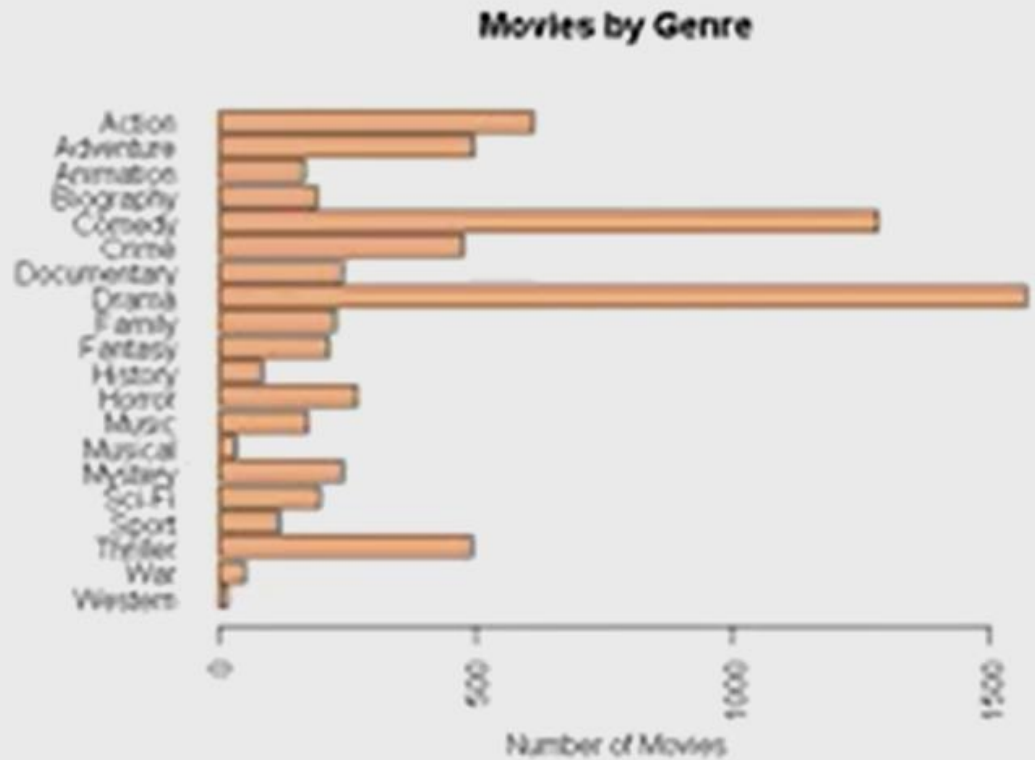
- **Purpose**: To display the frequency or proportion of different categories.
- **Description**: Each bar represents a category, and the height or length of the bar indicates the count or percentage of data in that category.
- **Example**: Showing the number of people in different age groups.

Horizontal Bar Chart

Many categories

Long names

Easier to read



Cleveland Dot Plot

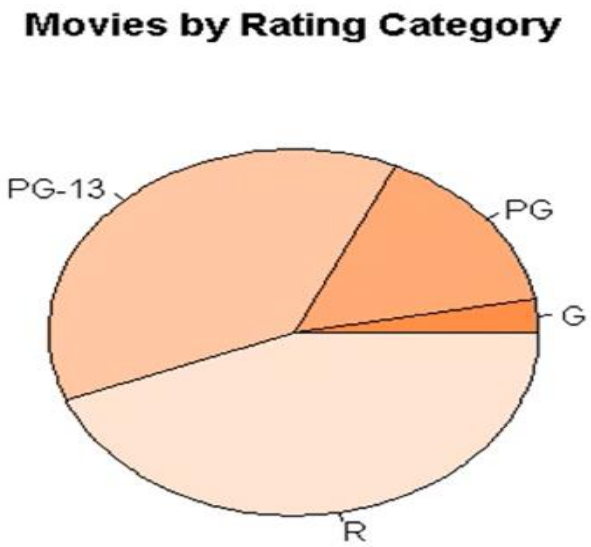
Data-to-ink ratio

Easier to read



Pie Chart

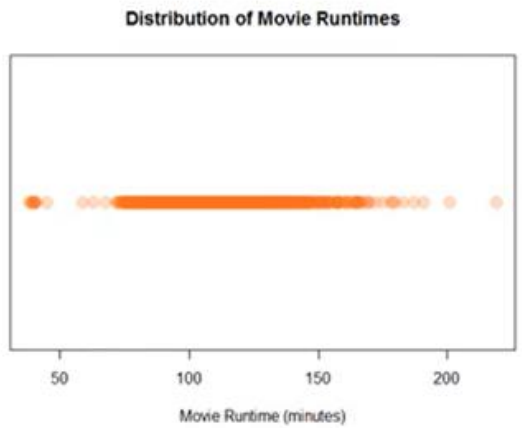
Proportion

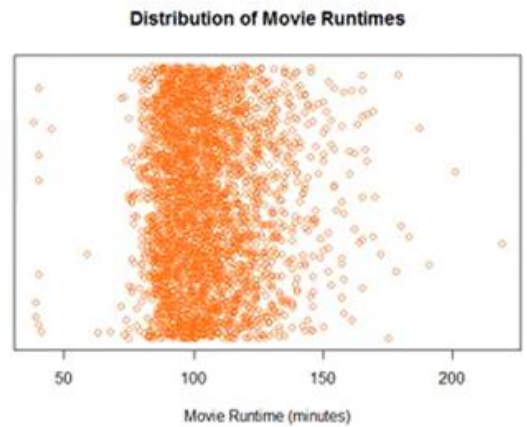


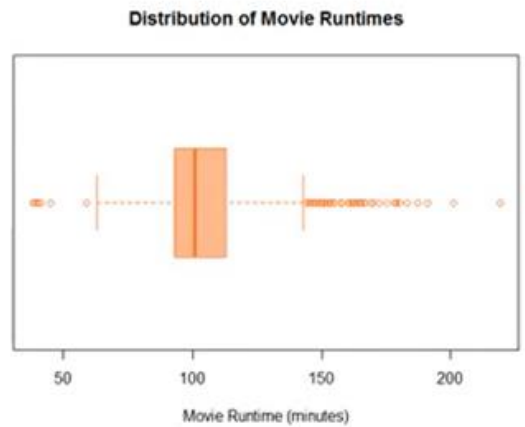
Pie Chart:

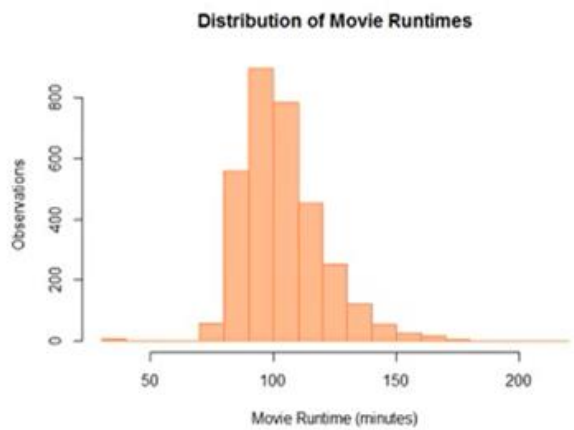
- **Purpose**: To show the proportion of categories as parts of a whole.
- **Description**: Each slice of the pie represents a category's proportion out of 100%.
- **Example**: Visualizing the market share of different brands in a market.

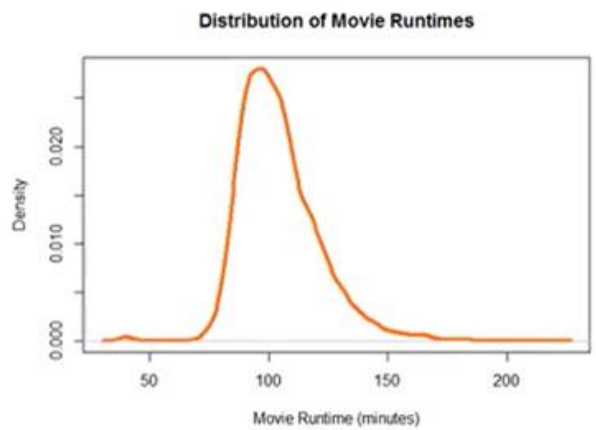
Visualizing One Numeric Variable











Numerical Univariate Visualization

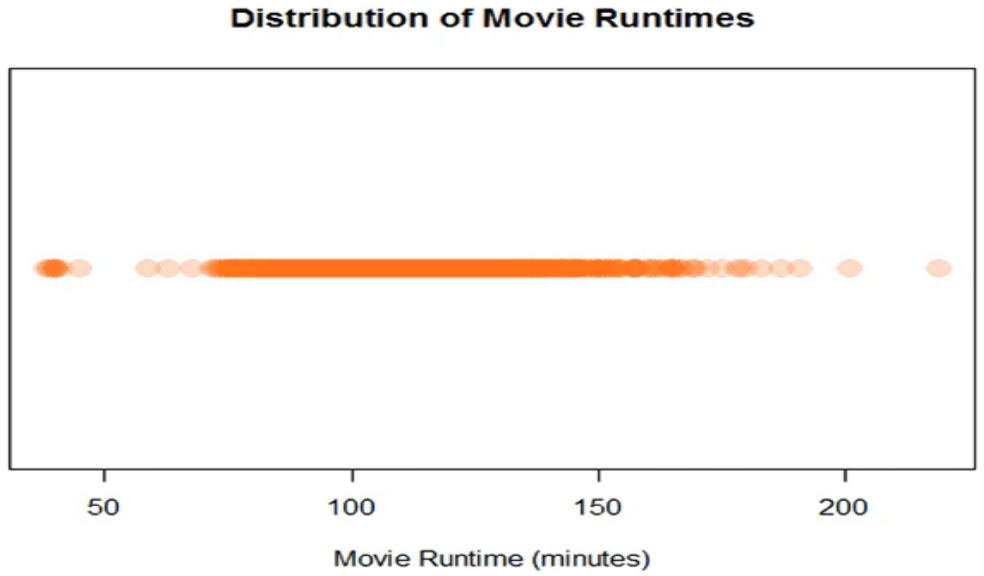
Numerical Univariate Visualization

Dot Plot

Location

Simple

Small size



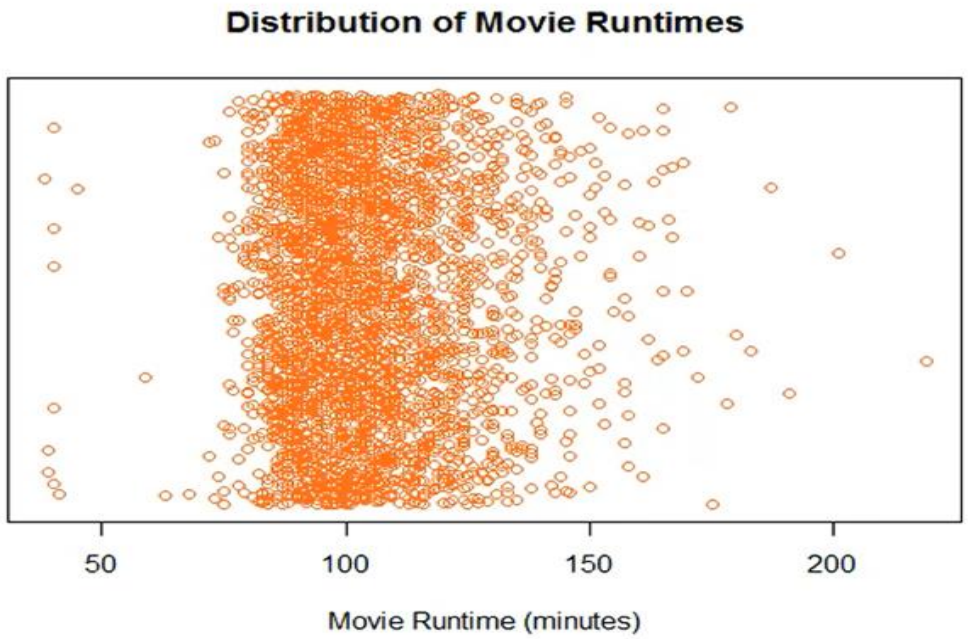
- **Purpose**: To display the distribution of a numerical variable and the frequency of individual values, making it easy to see the data's spread and identify patterns or clusters.
- **Description**: A dot plot uses dots to represent individual data points along a single axis. Each dot represents a data point, and multiple dots stacked vertically indicate higher frequencies of that value. It is a simple and effective way to visualize the distribution and frequency of data, especially for smaller datasets.
- **Example**: Showing the number of students scoring specific grades on a test, where each dot represents one student's score.

Jitter Plot

Noise

Larger size

Discrete values



Purpose: To visualize the distribution of numerical variable when data points are overlapping or clustered, improving visibility and readability.

Description: When applied to univariate data, jittering introduces a small amount of random noise along the axis to prevent overlap of data points, making it easier to see the distribution and density of data. This technique is especially useful for categorical variables or when visualizing the frequency of values.

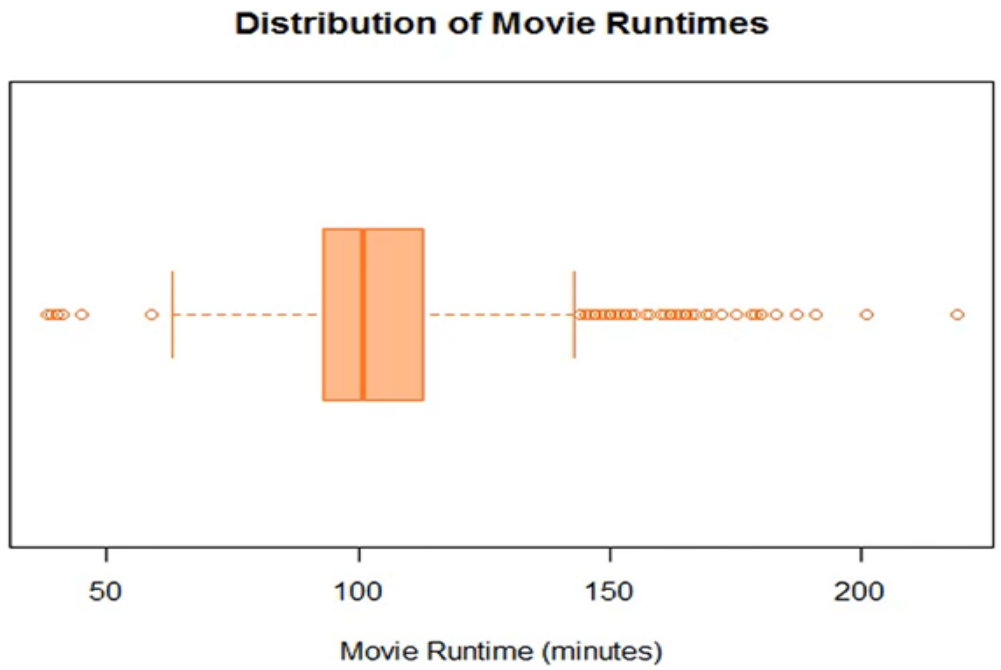
Example: Plotting the heights of students in a specific grade, where multiple students have the same height. Jittering helps in displaying these overlapping data points more clearly.

Box Plot

Location

Spread

Outliers



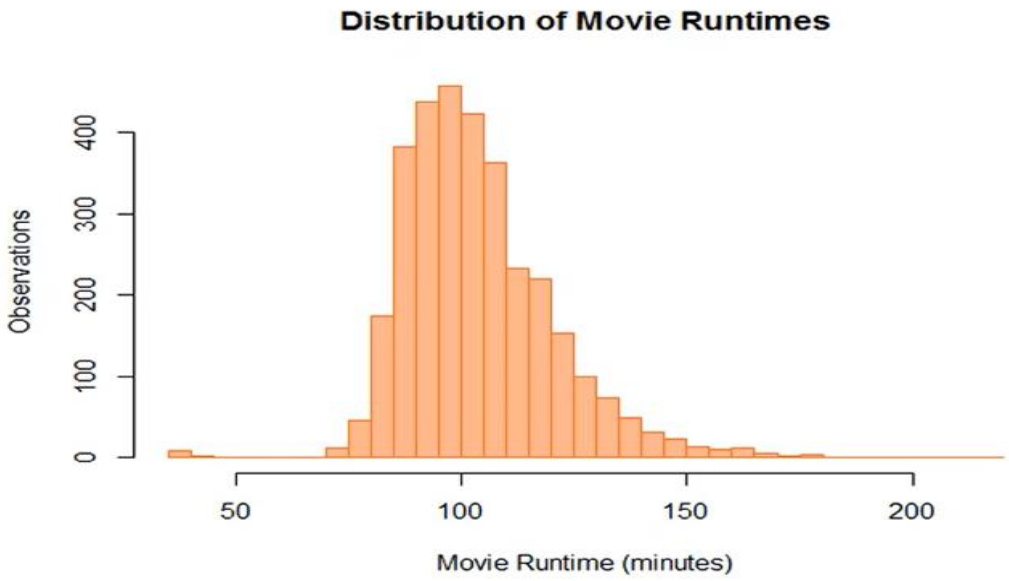
Box Plot (or Box-and-Whisker Plot):

- **Purpose**: To summarize the distribution of a numerical variable, including median, quartiles, and potential outliers.
- **Description**: Box plots display the minimum, first quartile, median, third quartile, and maximum of a data set, along with potential outliers.
- **Example**: Comparing the income distribution across different age groups.

Histogram

Bins

Shape



Histogram:

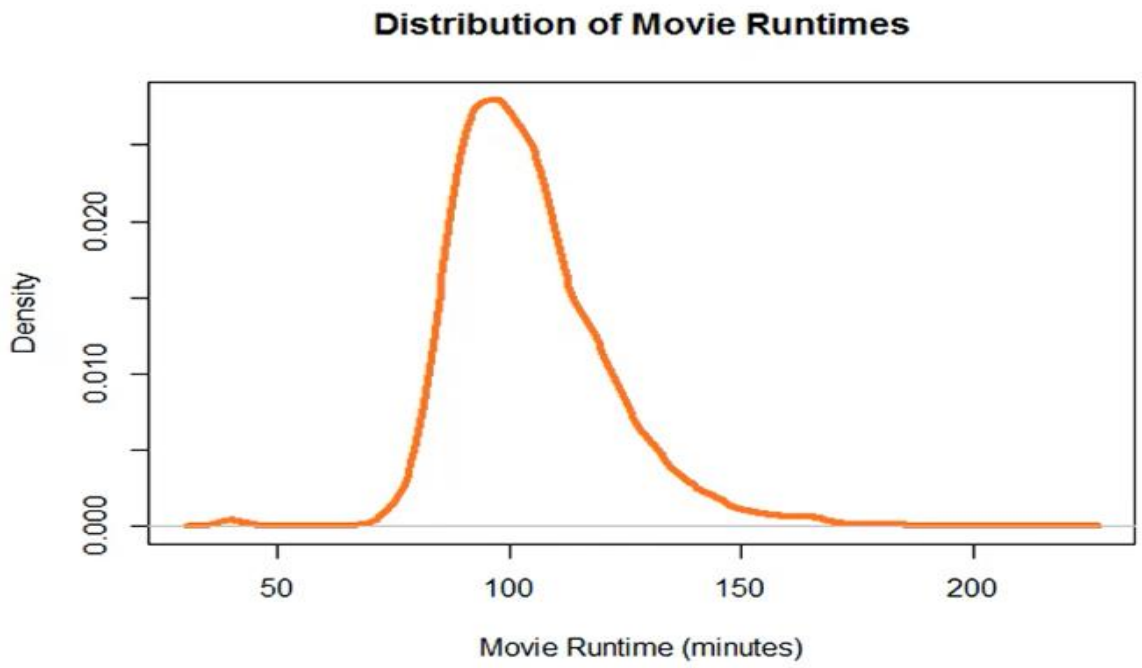
- **Purpose**: To show the distribution of a continuous numerical variable.
- **Description**: Histograms display data by grouping continuous data into bins or intervals. The height of each bar represents the frequency of data points within each interval.
- **Example**: Showing the distribution of heights in a population.

Density Plot

Shape

Smooth

Density



Density Plot:

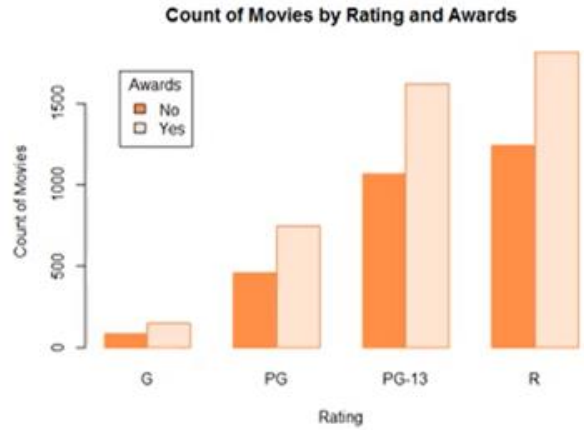
- **Purpose**: To show the distribution of a numerical variable, similar to a histogram but with a smoother curve.
- **Description**: Density plots use a continuous line to estimate the probability density function of a numerical variable.
- **Example**: Visualizing the distribution of test scores for a class.

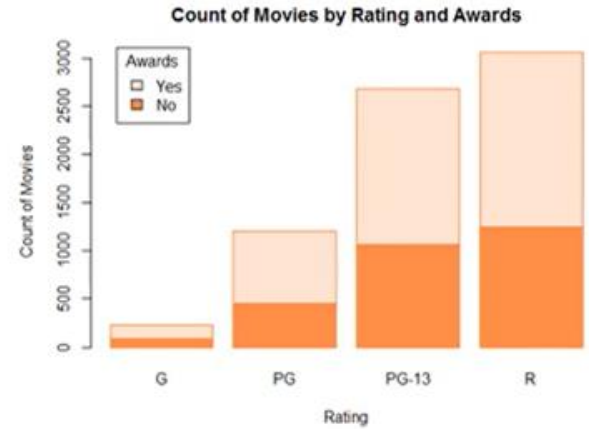
Qualitative Bivariate Visualization

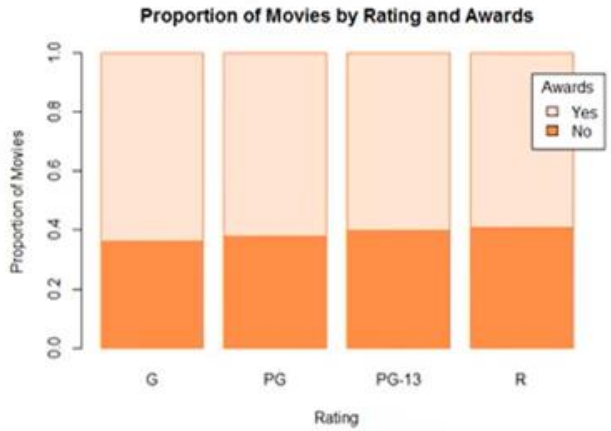
Qualitative Bivariate Visualization

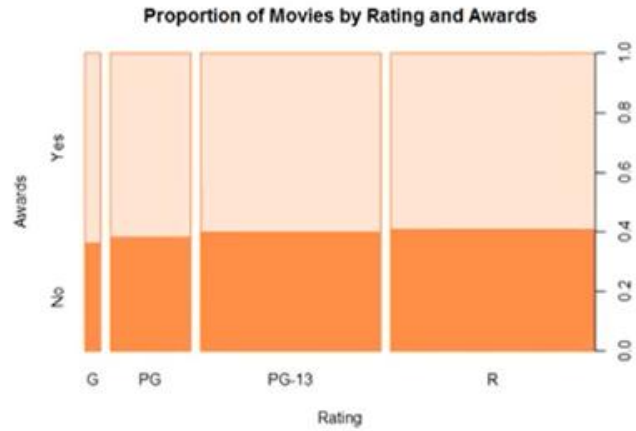
Visualizing Two Categorical Variables

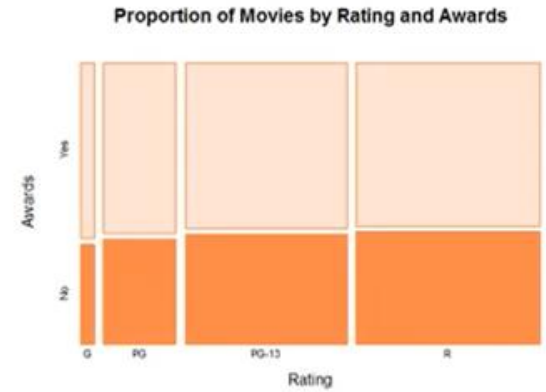
Movies by Genre and Rating					
Genre	G	PG	PG-13	R	Total
Action	2	70	311	229	612
Adventure	44	179	209	64	496
Animation	43	111	8	6	168
Comedy	45	258	472	506	1218
Drama	12	136	586	836	1570
Family	38	181	10	1	230
...	...	...	...	...	...
Total	230	1207	2686	3058	7181











Qualitative Bivariate Analysis

Joint frequency

Movies by Genre and Rating				
Genre	G	PG	PG-13	R
Action	2	70	311	229
Adventure	44	179	209	64
Animation	43	111	8	6
Comedy	45	258	472	506
Drama	12	136	586	836
Family	38	181	10	1
...	...	...	...	...

Qualitative Bivariate Analysis

Joint frequency

Marginal frequency

Movies by Genre and Rating					
Genre	G	PG	PG-13	R	Total
Action	2	70	311	229	612
Adventure	44	179	209	64	496
Animation	43	111	8	6	168
Comedy	45	258	472	506	1218
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Family	38	181	10	1	230
...	...	...	...	...	...
Total	230	1207	2686	3058	7181

Qualitative Bivariate Analysis

Joint frequency

Marginal frequency

Relative frequency

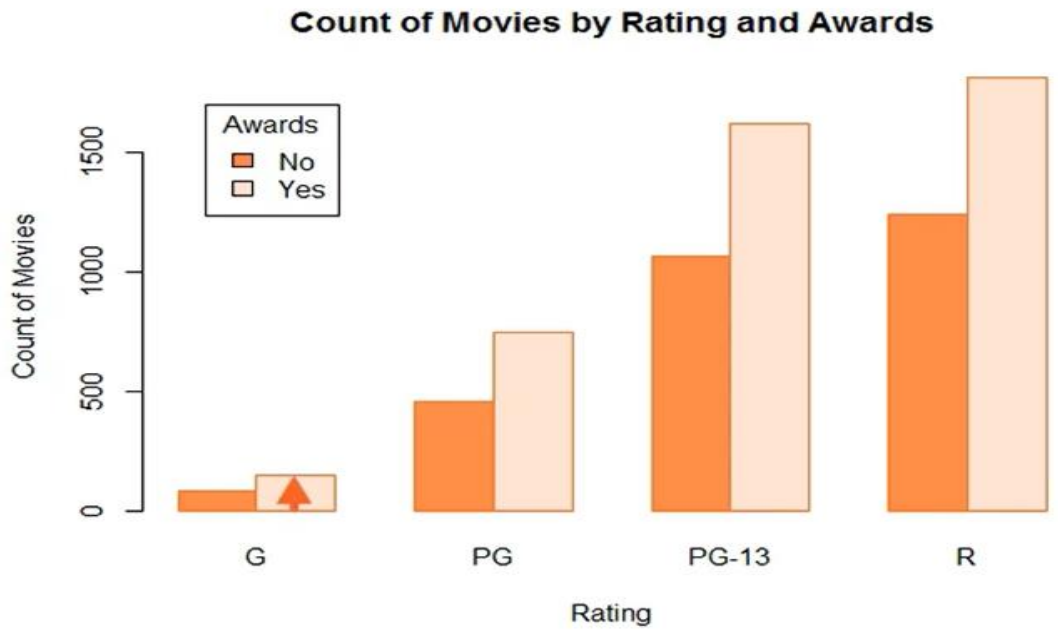
Movies by Genre and Rating					
Genre	G	PG	PG-13	R	Total
Action	0.001	0.010	0.043	0.032	0.086
Adventure	0.006	0.025	0.029	0.009	0.069
Animation	0.006	0.015	0.001	0.001	0.023
Comedy	0.006	0.036	0.066	0.070	0.170
Drama	0.002	0.019	0.082	0.116	0.219
Family	0.005	0.025	0.001	0.001	0.033
...	...	...	...	...	...
Total	0.032	0.168	0.374	0.426	1.000

Grouped Frequency Bar Chart

Joint frequency

Across categories

Within categories



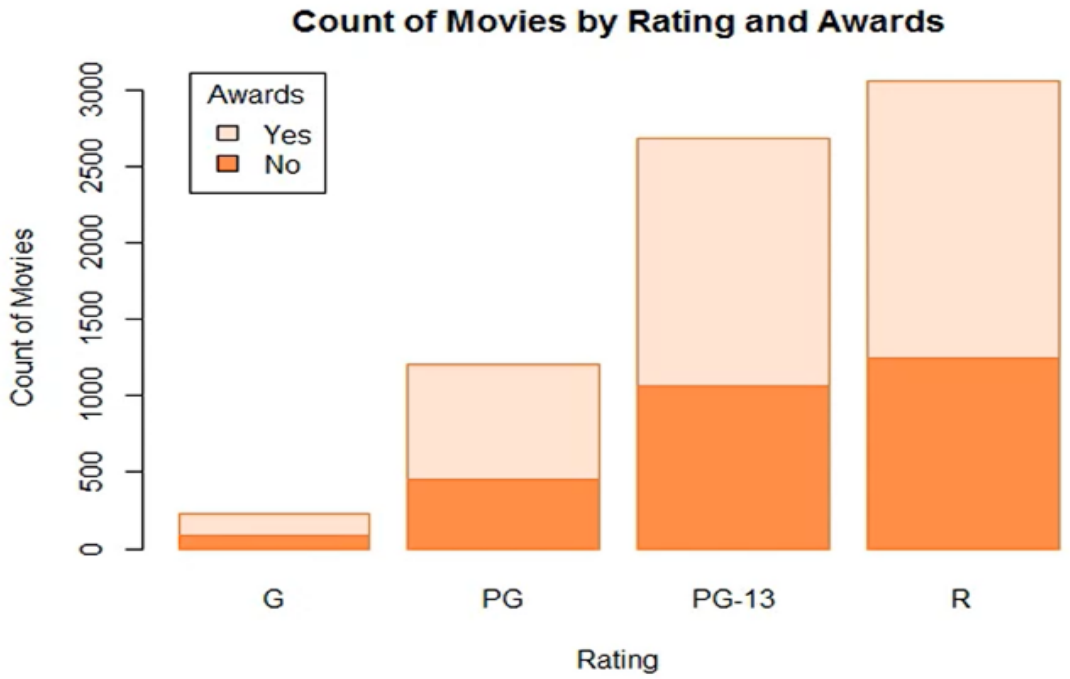
- **Purpose**: To compare multiple categories across different groups, showing the distribution of values for each category within each group.
- **Description**: A grouped bar chart displays bars for different categories side by side within each group. Each group of bars represents a different category or sub-category, allowing for easy comparison between groups and categories.
- **Example**: Comparing sales figures for different products across various regions, where each group represents a region and each bar within the group represents a different product's sales.

Stacked Frequency Bar Chart

Marginal frequency

Joint frequency

Part of whole



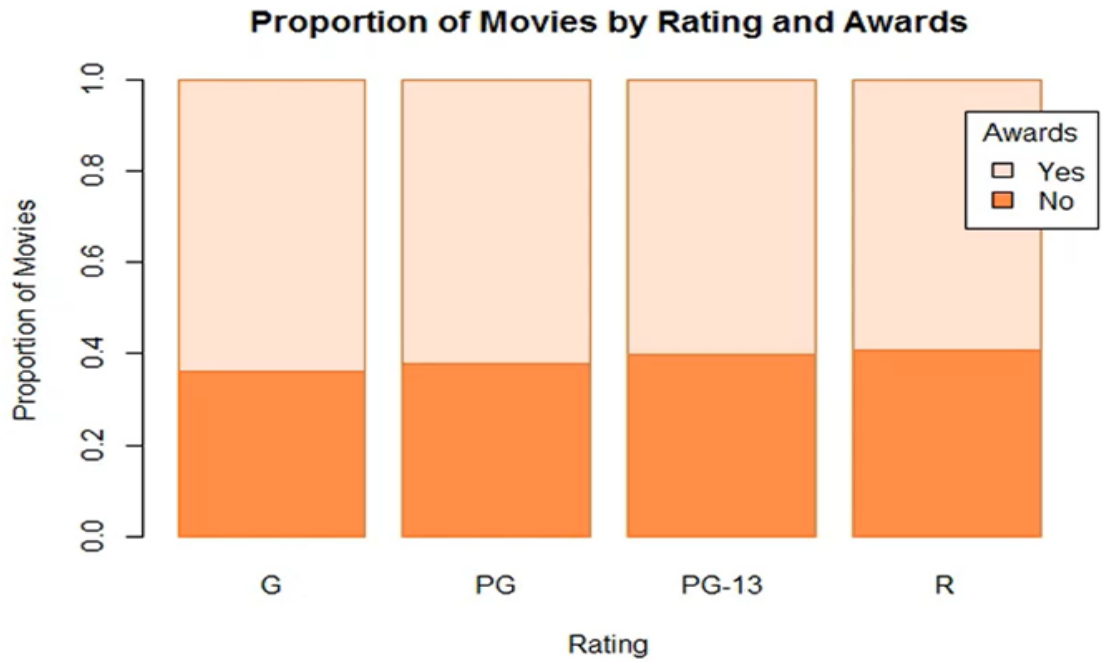
• **Grouped Bar Chart**: Displays bars for each category side-by-side within each group. Each group of bars represents different categories or sub-categories, allowing for easy comparison across both groups and categories.

• **Stacked Bar Chart**: Displays bars divided into segments, where each segment represents a sub-category. The segments stack on top of each other, showing the total value for each group and how different sub-categories contribute to that total.

100% Stacked Frequency Bar Chart

Relative frequency

One variable



- **Stacked Frequency Bar Chart** focuses on visualizing frequency counts and how different sub-categories contribute to the overall frequency within each category.
- **Stacked Bar Chart** is more general and can be used to show the breakdown of any aggregated values, not limited to frequencies, and highlights the proportion of sub-categories within each category.

Spine Plot

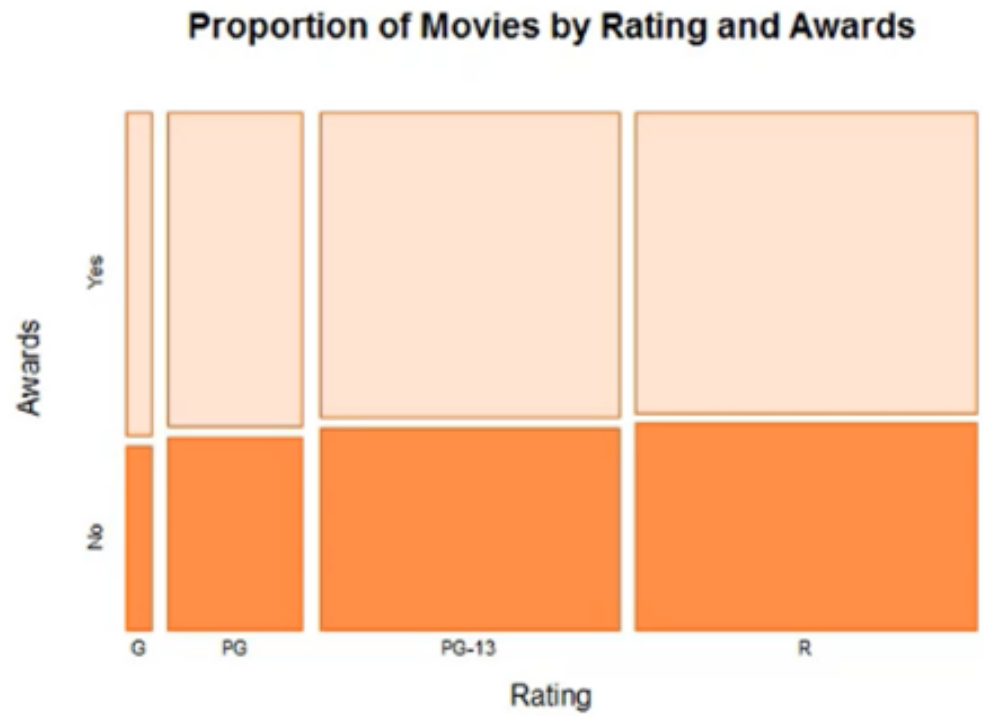
Relative frequency

Both variables



Mosaic Plot

Clean aesthetics

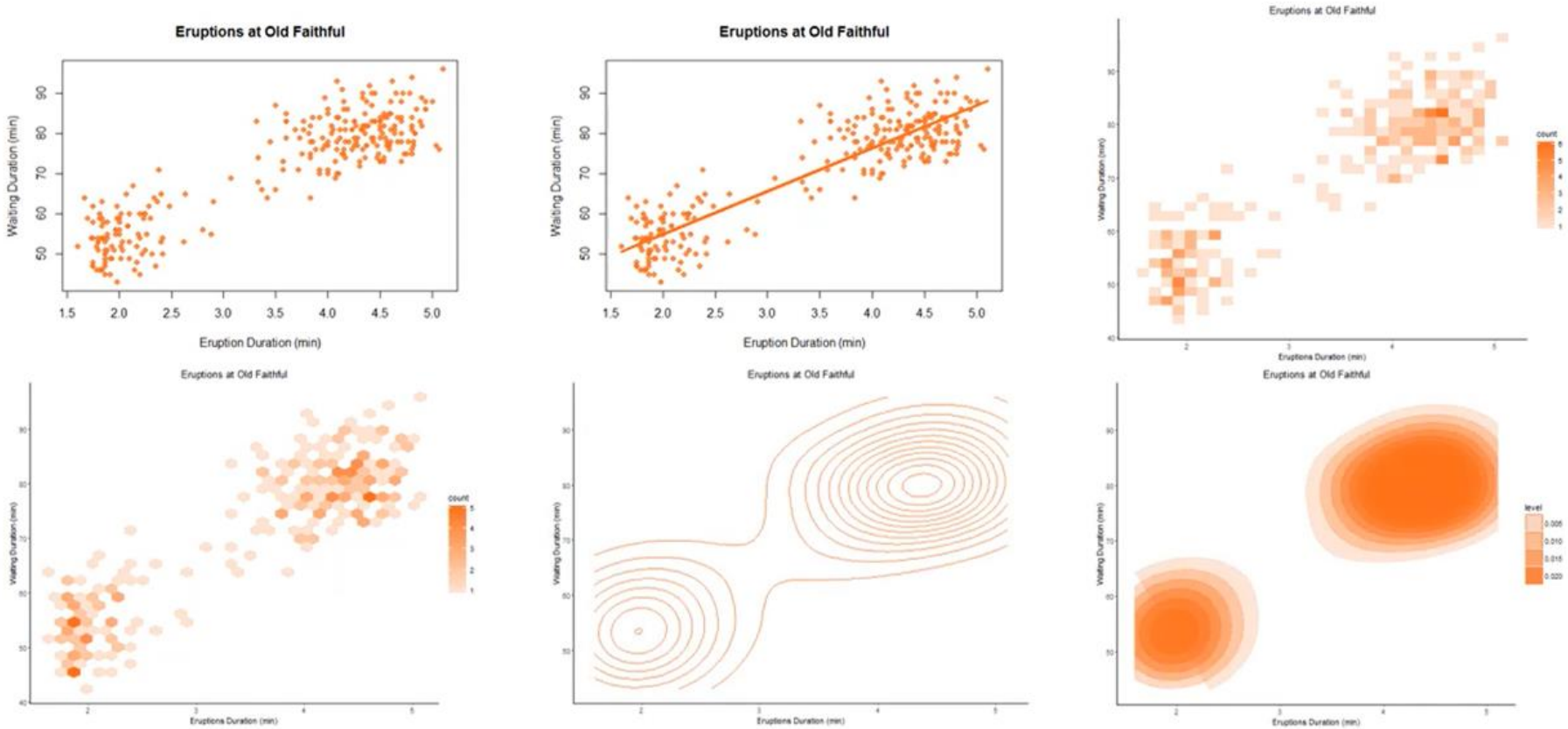


Quantitative Bivariate Visualization



Quantitative Bivariate Visualization

Visualizing Two Numeric Variables

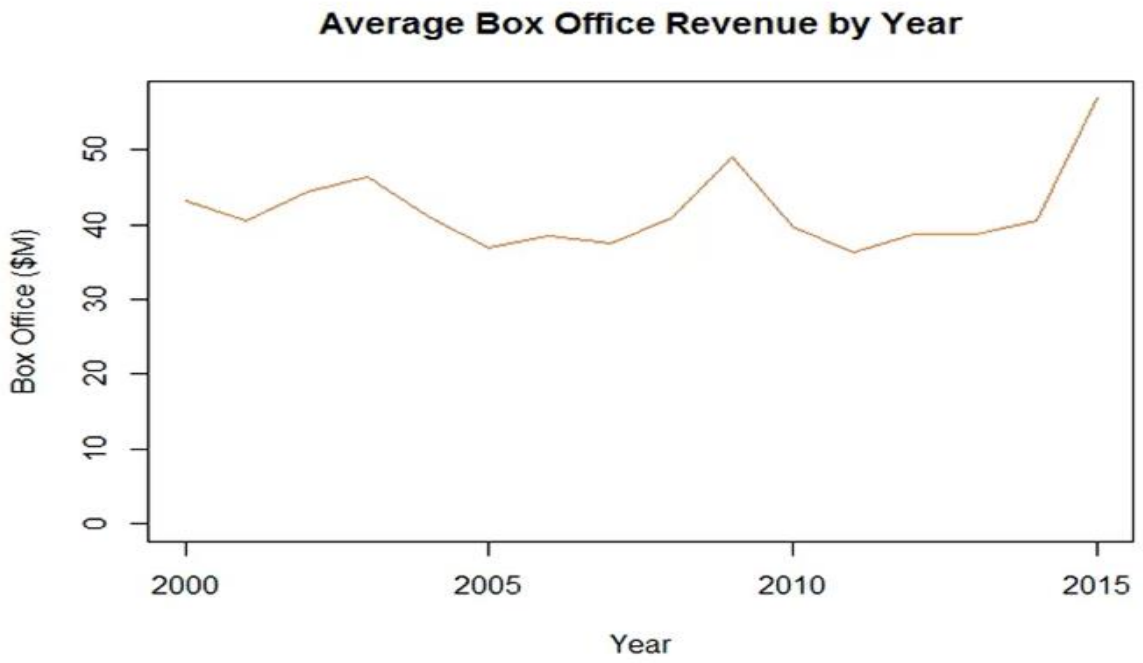


Let us explore some of the important Charts used for Two numeric variable

Line Chart

Change over time

Rate of change

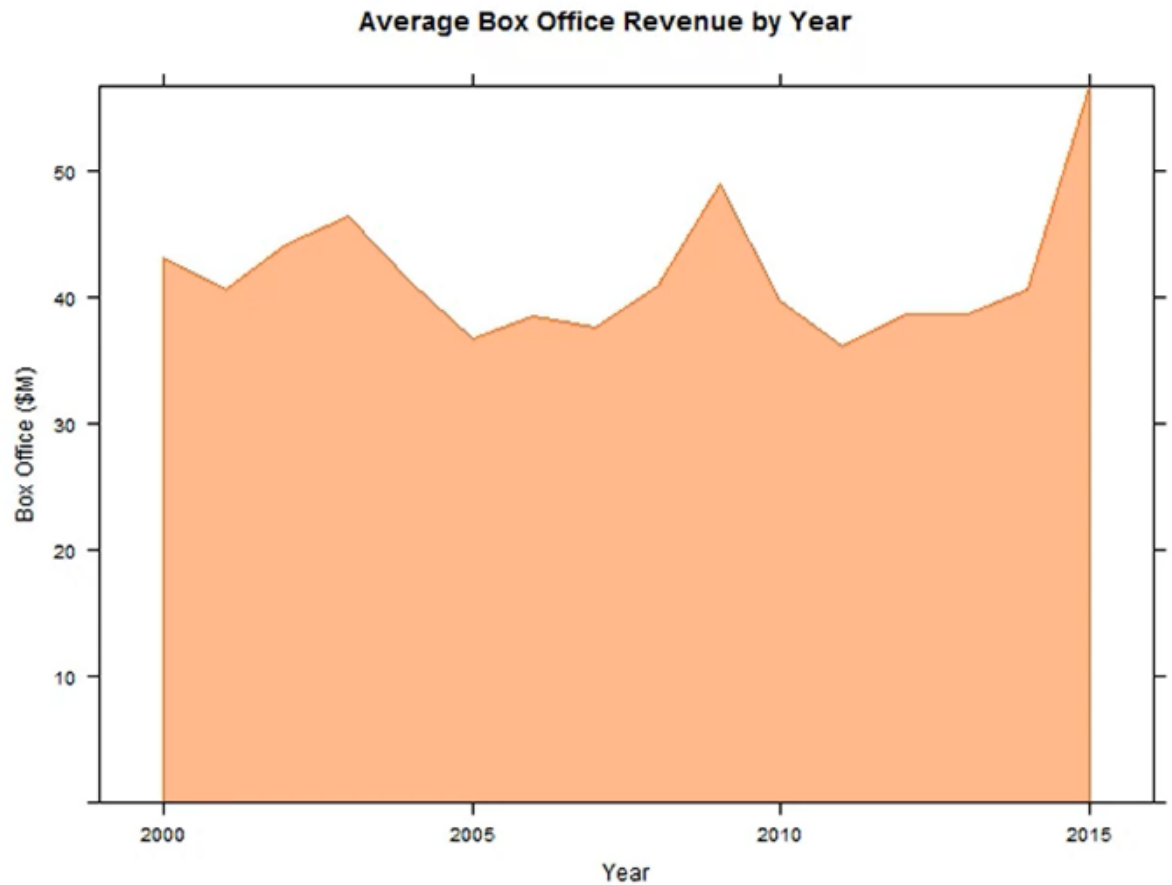


Line Chart:
Purpose: To show trends over time or sequential data.
Description: Points are plotted and connected with a line, showing how one numerical variable changes in relation to another (often time).
Example: Visualizing the stock price of a company over time

Area Chart

Change over time

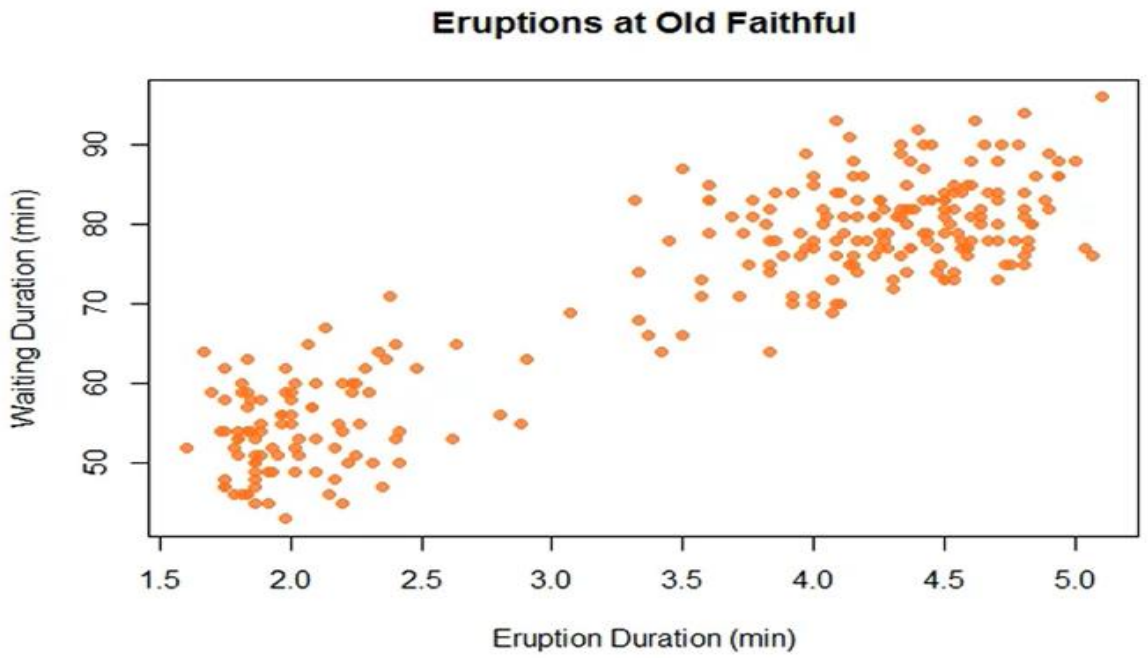
Volume / summation



Scatterplot

Correlation

Shape



Scatter Plot:

Purpose: To show the relationship between two numerical variables.

Description: Each point represents an observation, with the x-axis and y-axis representing the two variables.

Example: Analyzing the relationship between hours studied and exam scores.

Binned Frequency Heatmap

Joint frequency

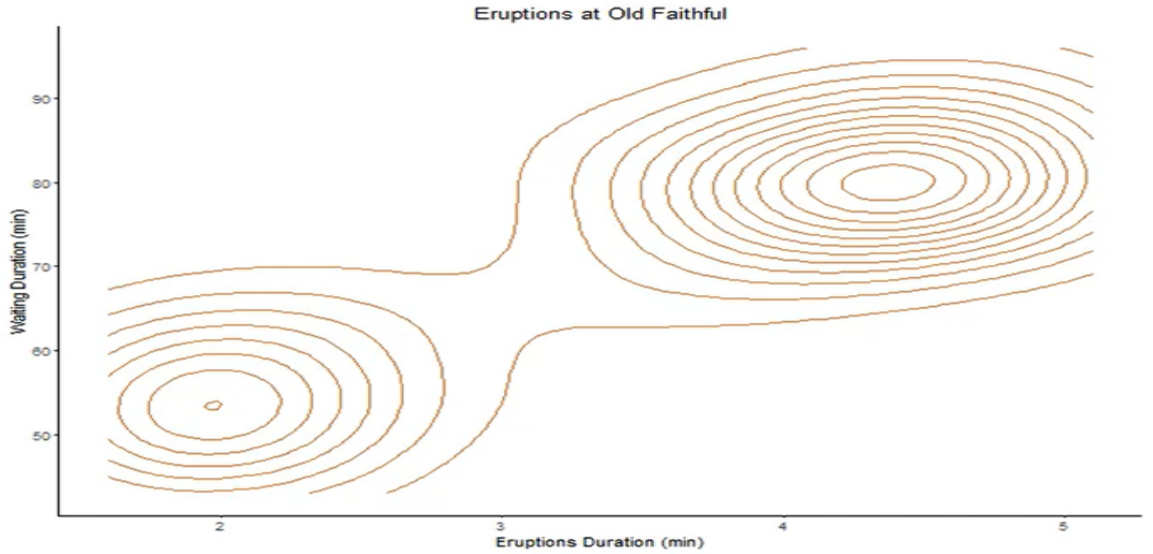
Large data sets



Contour Plot

Contour lines

Joint density



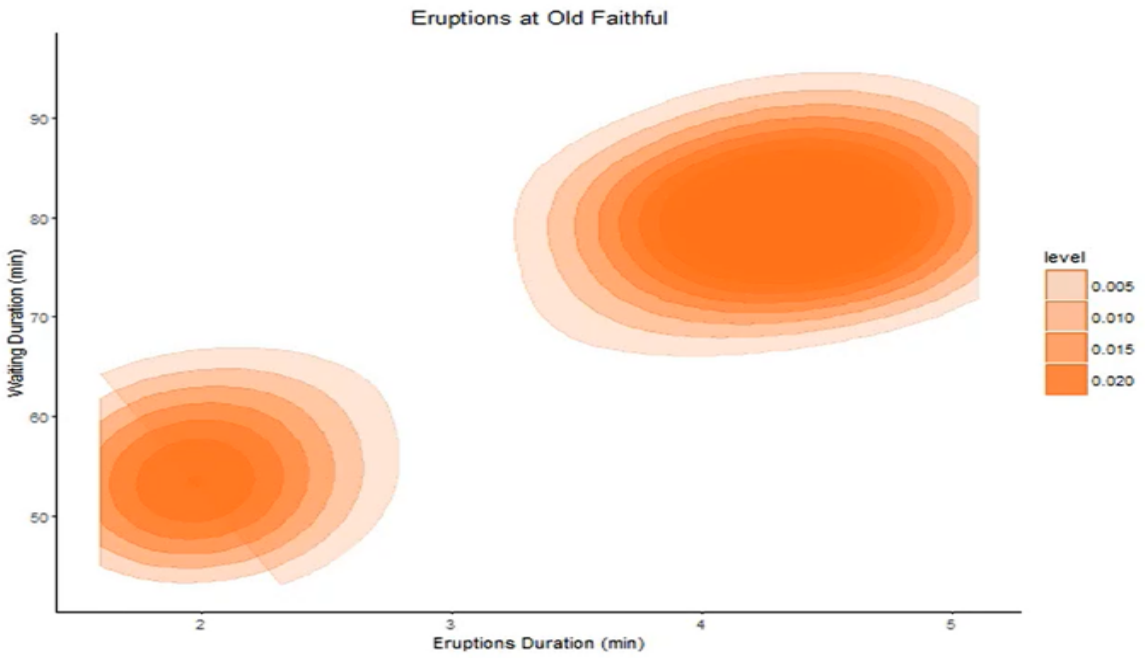
Purpose: To visualize the relationship between three numerical variables, showing how one variable changes with the other two, using contour lines.

- **Description**: A contour plot displays lines of constant values, called contour lines,
- which represent levels of a third variable across a two-dimensional plane. The density and shape of these lines indicate the gradient and distribution of the variable.
- **Example**: Mapping elevation levels on a geographical terrain to show the contours of hills and valleys.

Level Plot

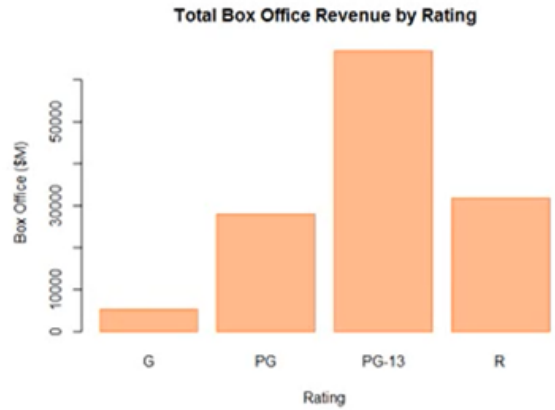
Colored levels

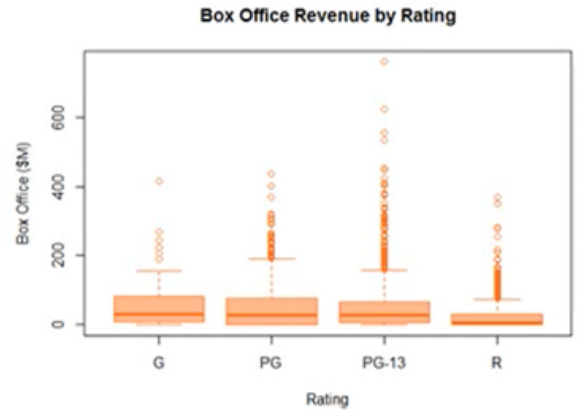
Lower data-to-ink

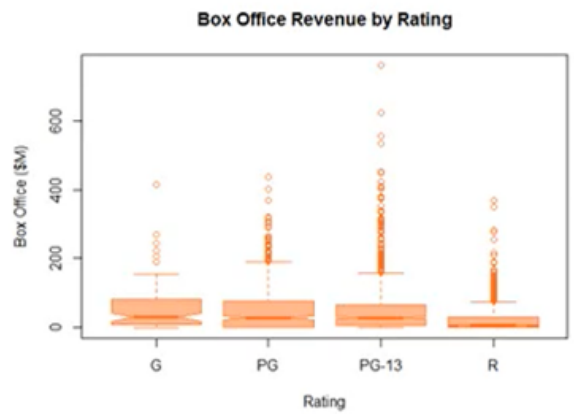


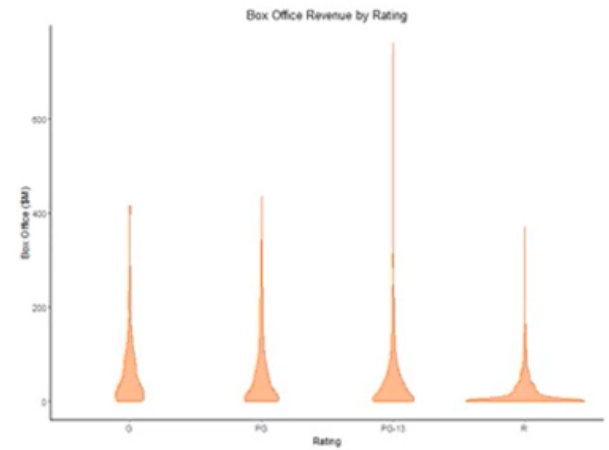
A contour plot displays lines (contours) that connect points of equal value, showing the distribution of a variable in two dimensions. A level plot, on the other hand, fills the space between contour lines with color bands to represent different levels of the variable, providing a more visually intuitive gradient.

Visualizing Both a Categorical and a Numeric Variable









Qualitative and Quantitative Bivariate Analysis

Aggregate

Grouped

Comparison



Bar Chart (with numerical data):

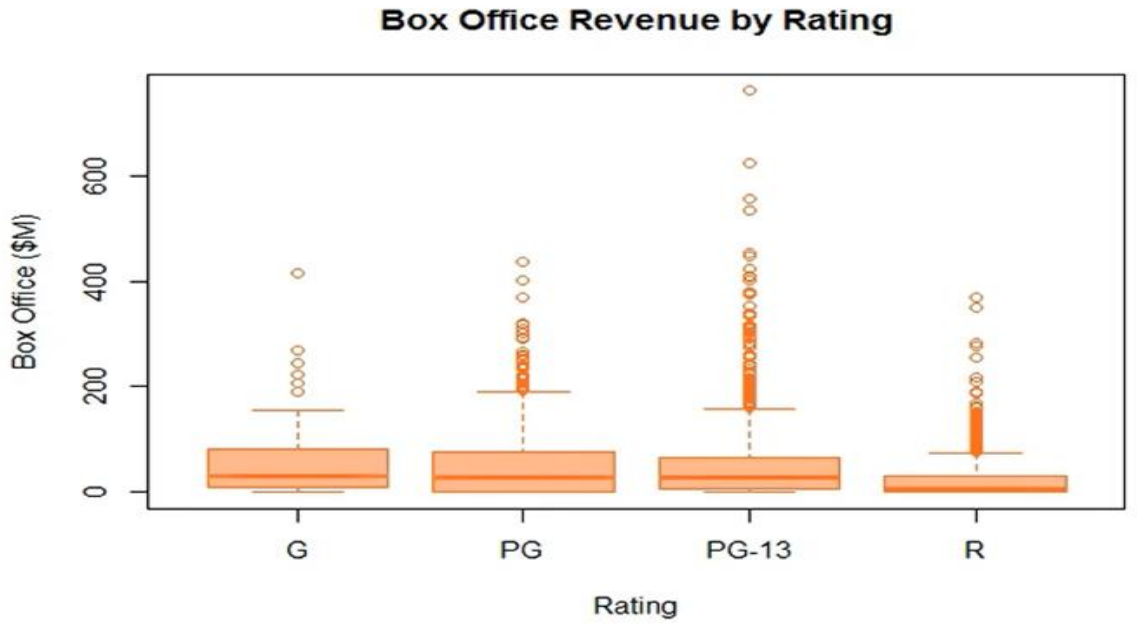
- **Purpose**: To compare numerical values across different categories.
- **Description**: Bars represent the average or total of a numerical variable for each category.
- **Example**: Showing the average sales figures for different regions.

Bivariate Box Plot

Location

Spread

Outliers



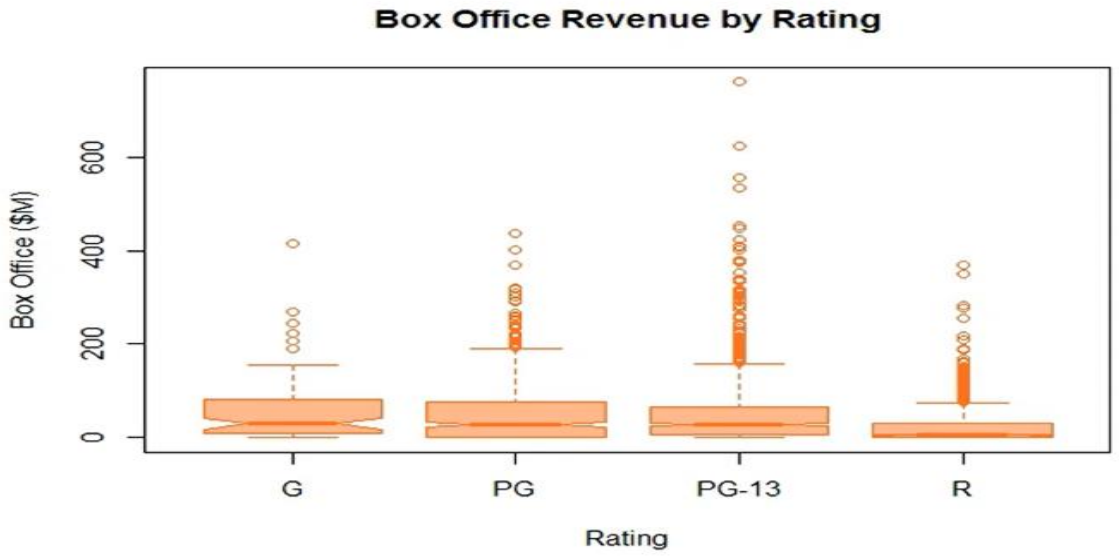
Box Plot (for different categories):

- **Purpose**: To compare the distribution of a numerical variable across different categories.
- **Description**: Box plots for each category are displayed side-by-side.
- **Example**: Comparing salaries across different job titles.

Notched Box Plot

Notches

Confidence interval



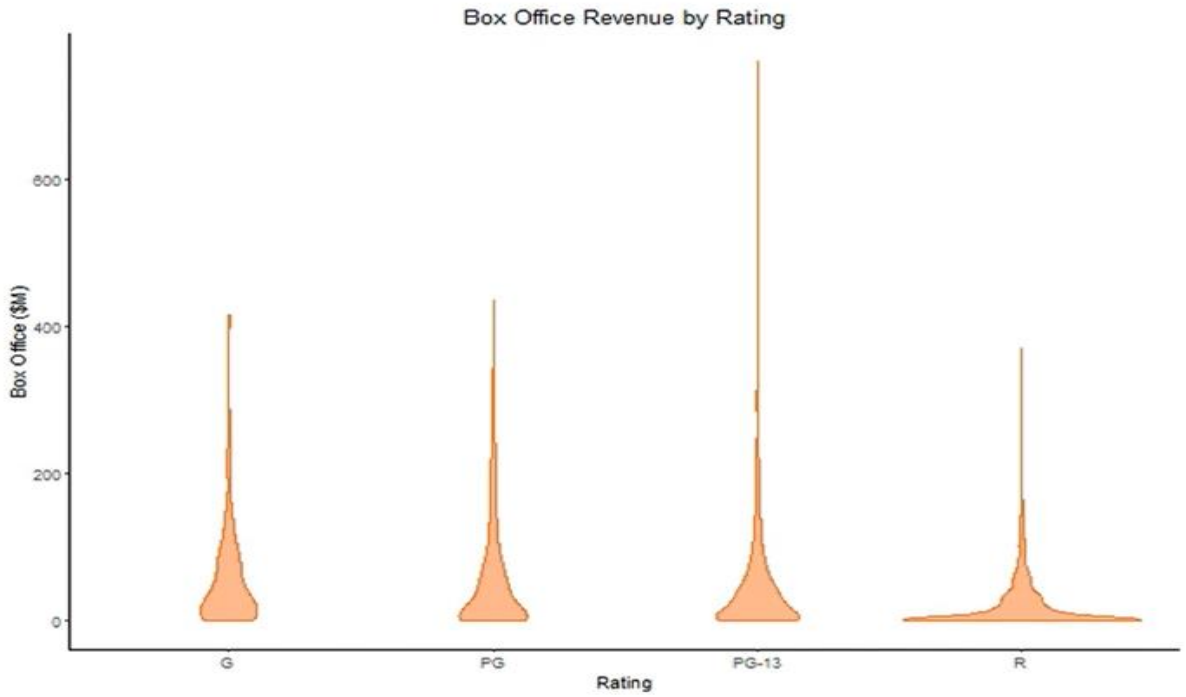
Notched Box Plot

- **Purpose**: To compare the medians of different groups and provide a visual indication of the statistical significance of these differences.
- **Description**: A notched box plot is a variation of the standard box plot that includes notches around the median, representing a confidence interval. If the notches between two groups do not overlap, it suggests a statistically significant difference between their medians.
- **Example**: Comparing the median test scores of students from three different schools to determine if there are significant differences in performance.

Bivariate Violin Plot

Density

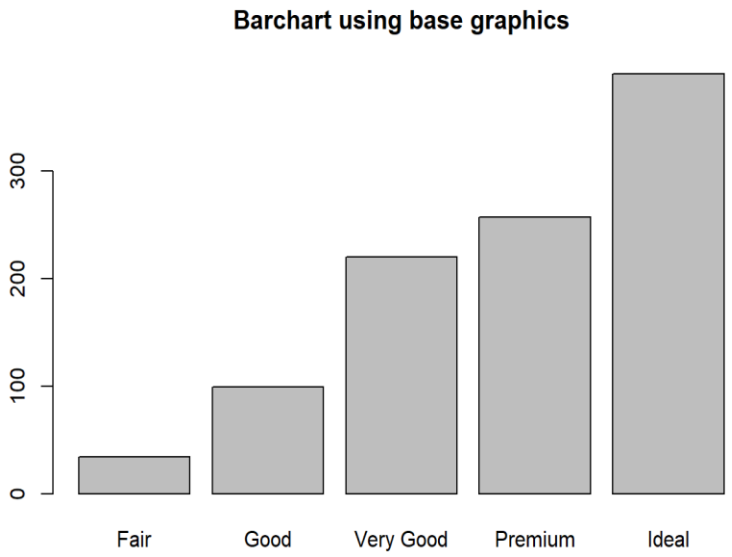
Shape



Violin Plot:

- **Purpose**: To visualize the distribution of a numerical variable for different categories, combining the box plot and density plot.
- **Description**: Similar to a box plot, but also shows the probability density of the data at different values.
- **Example**: Displaying the distribution of test scores by gender.

Inferences from Visualization



Bar Chart

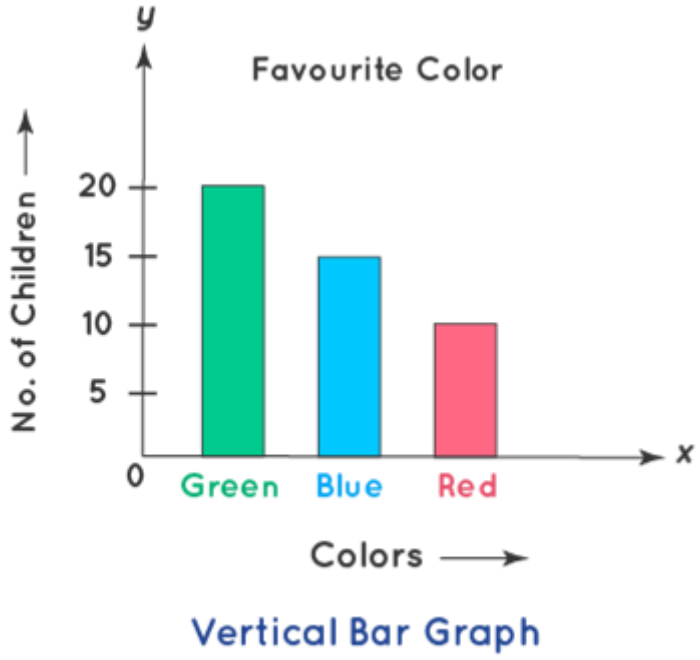
- This is one of the most common types of data visualization tools.
- Column chart is to track data sets over time.
- A column chart will include data labels along the horizontal (X) axis with measured metrics or values presented on the vertical (Y) axis, also known as the left side of the chart. The Y-axis will normally start at 0 and go as high as the largest measurement you're tracking.

Use bar charts to do the following:

- Compare counts by categories.
- Display a variable function (sum, average, standard deviation) by categories.
- Understand relationships between categorical variables.

Vertical Bar Graphs

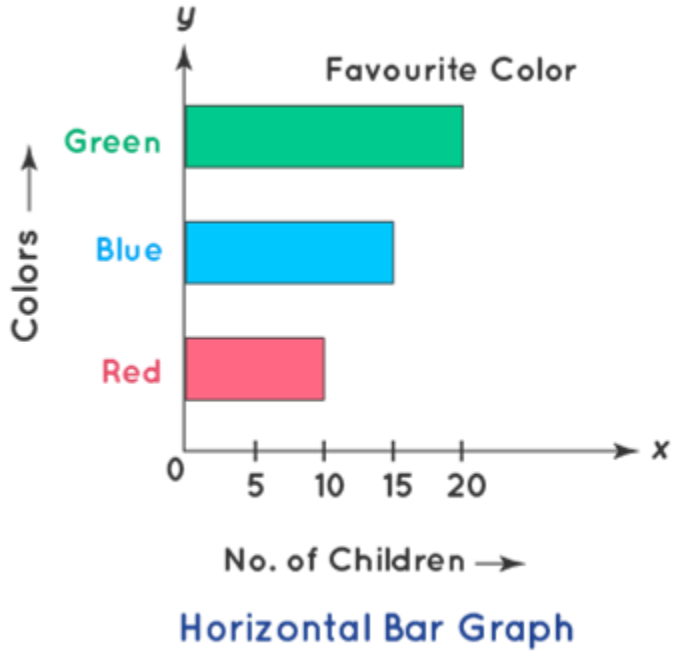
When the given data is represented vertically in a graph or chart with the help of rectangular bars that show the measure of data, such graphs are known as vertical bar graphs. The rectangular bars are vertically drawn on the x-axis, and the y-axis shows the value of the height of the rectangular bars which represents the quantity of the variables written on the x-axis.



Vertical Bar Graph

Horizontal Bar Graphs

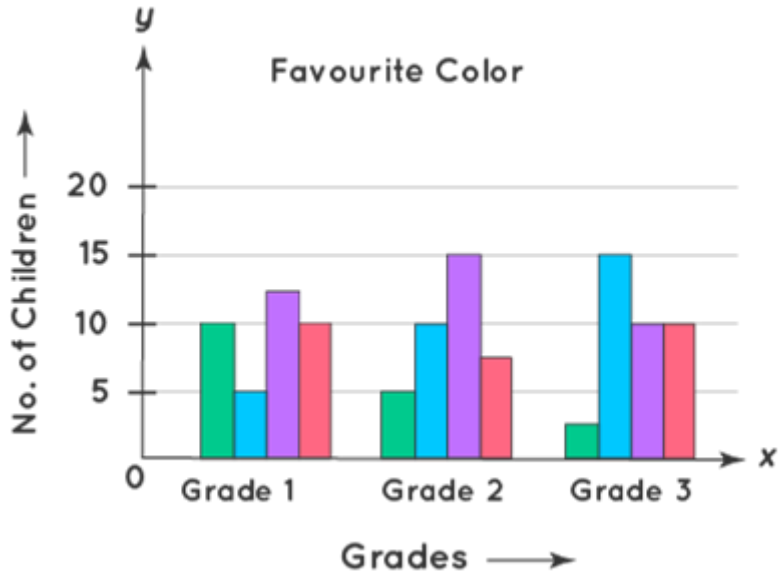
When the given data is represented horizontally by using rectangular bars that show the measure of data, such graphs are known as horizontal bar graphs. In this type, the variables or the categories of the data have to be written and then the rectangular bars are horizontally drawn on the y-axis and the x-axis shows the length of the bars equal to the values of different variables present in the data.



Horizontal Bar Graph

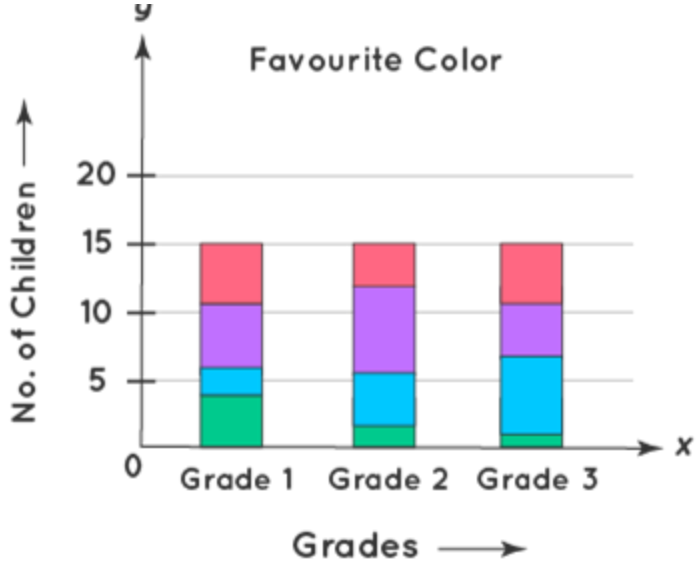
Grouped Bar Graph

The grouped bar graph is also referred the clustered bar graph. It is used to show the discrete value for two or more categorical data. In this, rectangular bars are grouped by position for levels of one categorical variable, with the same colours showing the secondary category level within each group. It can be shown both vertically and horizontally.
Observe the figure given below which shows different types of bar graphs.



Stacked Bar Graph

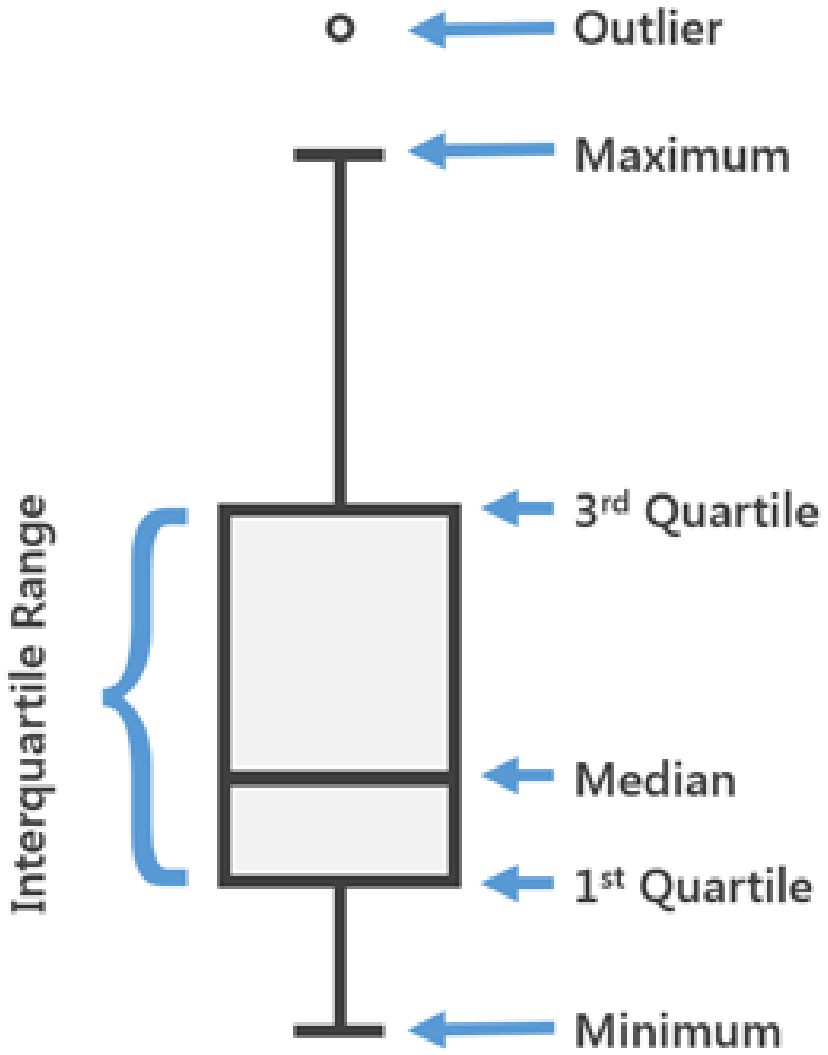
The stacked bar graph is also referred to as the composite bar graph. It divides the whole bar into different parts. In this, each part of a bar is represented using different colours to easily identify the different categories. It requires specific labelling to indicate the different parts of the bar. Thus, in a stacked bar graph every rectangular bar represents the whole, and each segment in the rectangular bar shows the different parts of the whole. It can be shown vertically or horizontally.



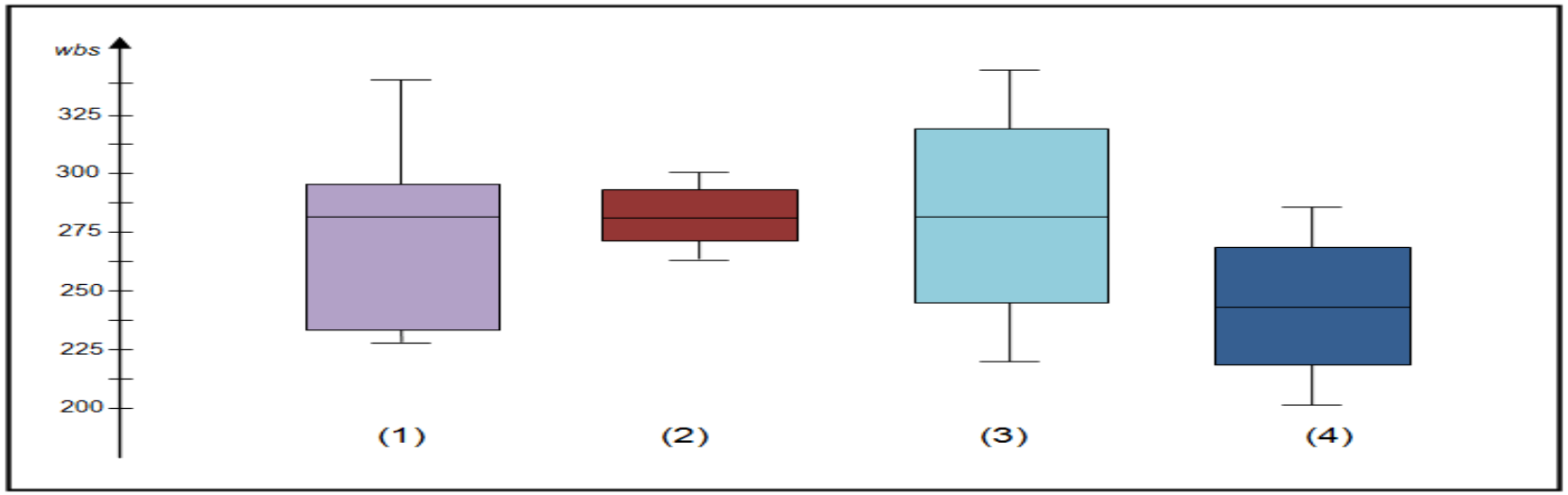
Box and Whisker Plot

- This plot can be used to obtain more **statistical details** about the data.
- The straight lines at the maximum and minimum are also called **whiskers**.
- Points that lie outside the whiskers will be considered as an outlier.
- The box plot also gives us a description of the **25th, 50th,75th quartiles**.
- With the help of a box plot, we can also determine the **Interquartile range(IQR)** where maximum details of the data will be present. Therefore, it can also give us a clear idea about the outliers in the dataset.

Box Plot


Outlier
Maximum
3rd Quartile
Interquartile Range
Median
1st Quartile
Minimum

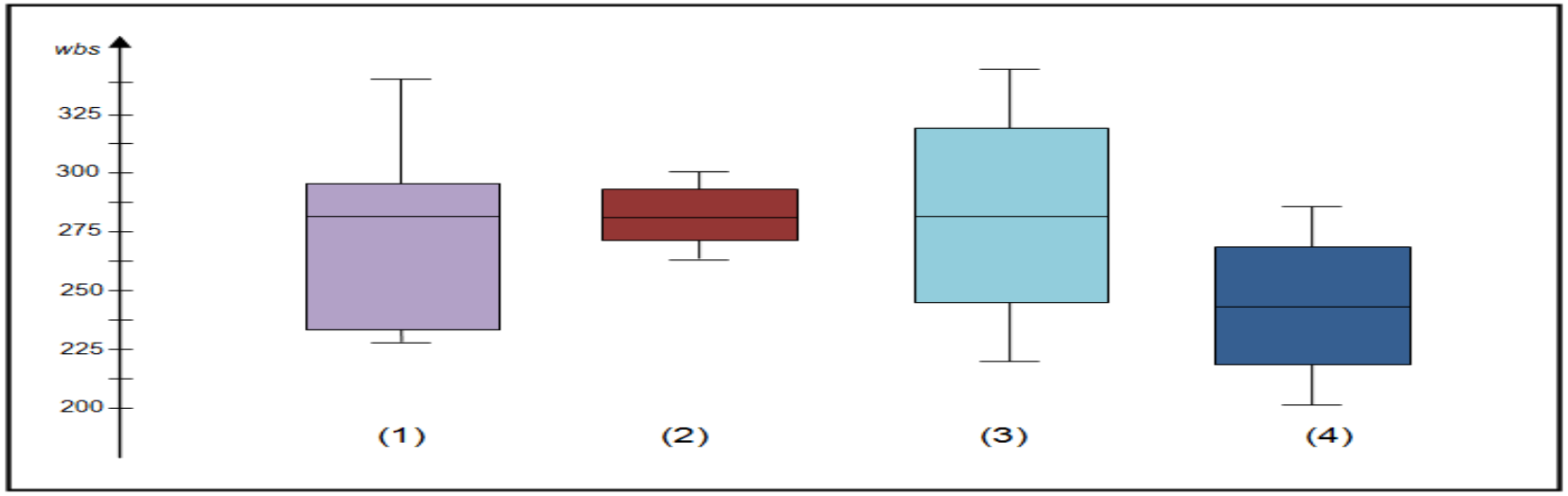
Inferences from Box Plot



The box plot is comparatively short – see example (2). This suggests that overall students have a high level of agreement with each other.

The box plot is comparatively tall – see examples (1) and (3). This suggests students hold quite different opinions about this aspect or sub-aspect.

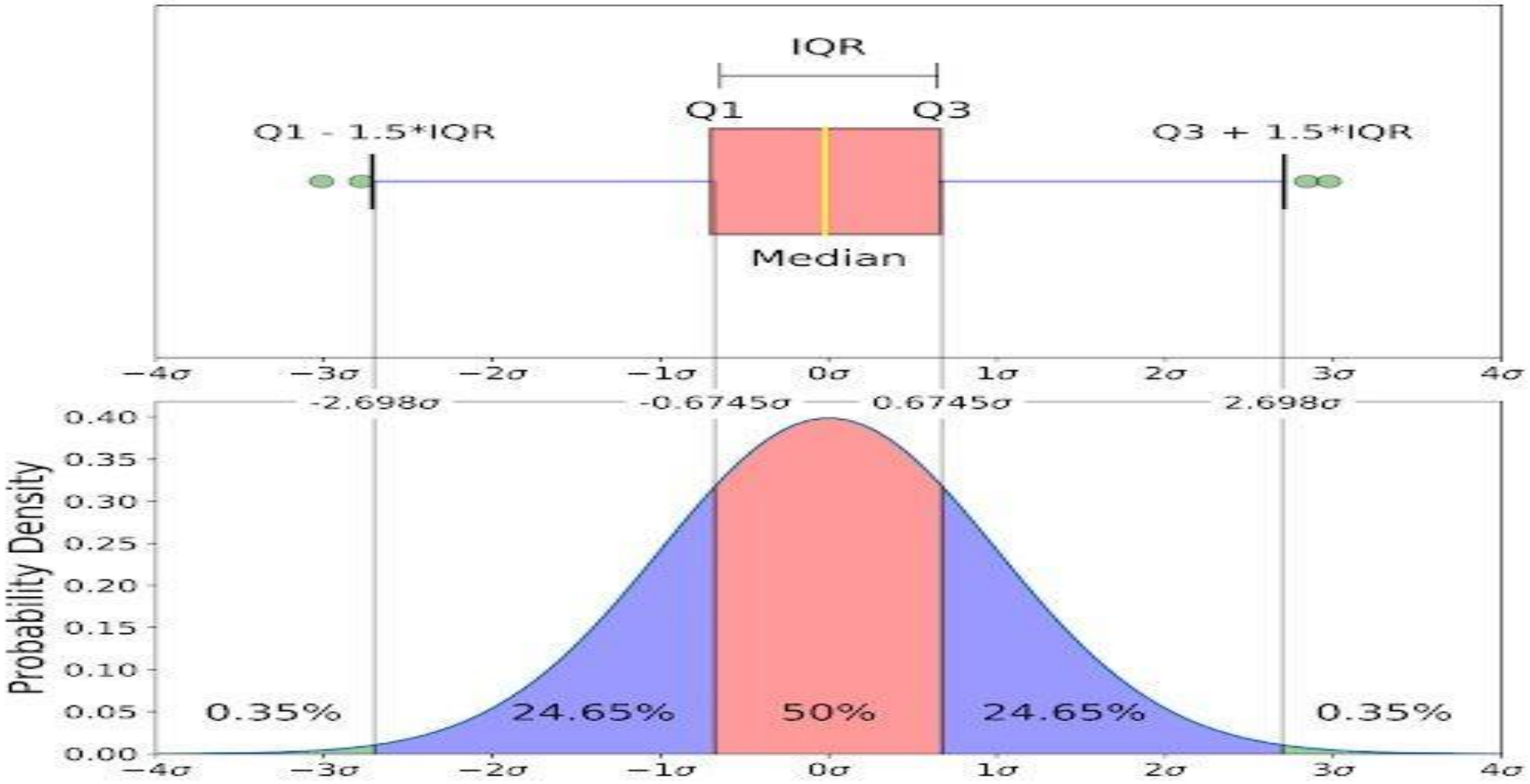
Inferences from Box Plot



One box plot is much higher or lower than another – compare (3) and (4) – This could suggest a difference between groups. For example, the box plot for boys may be lower or higher than the equivalent plot for girls. Follow this up by looking at the *Items at a Glance* reports.

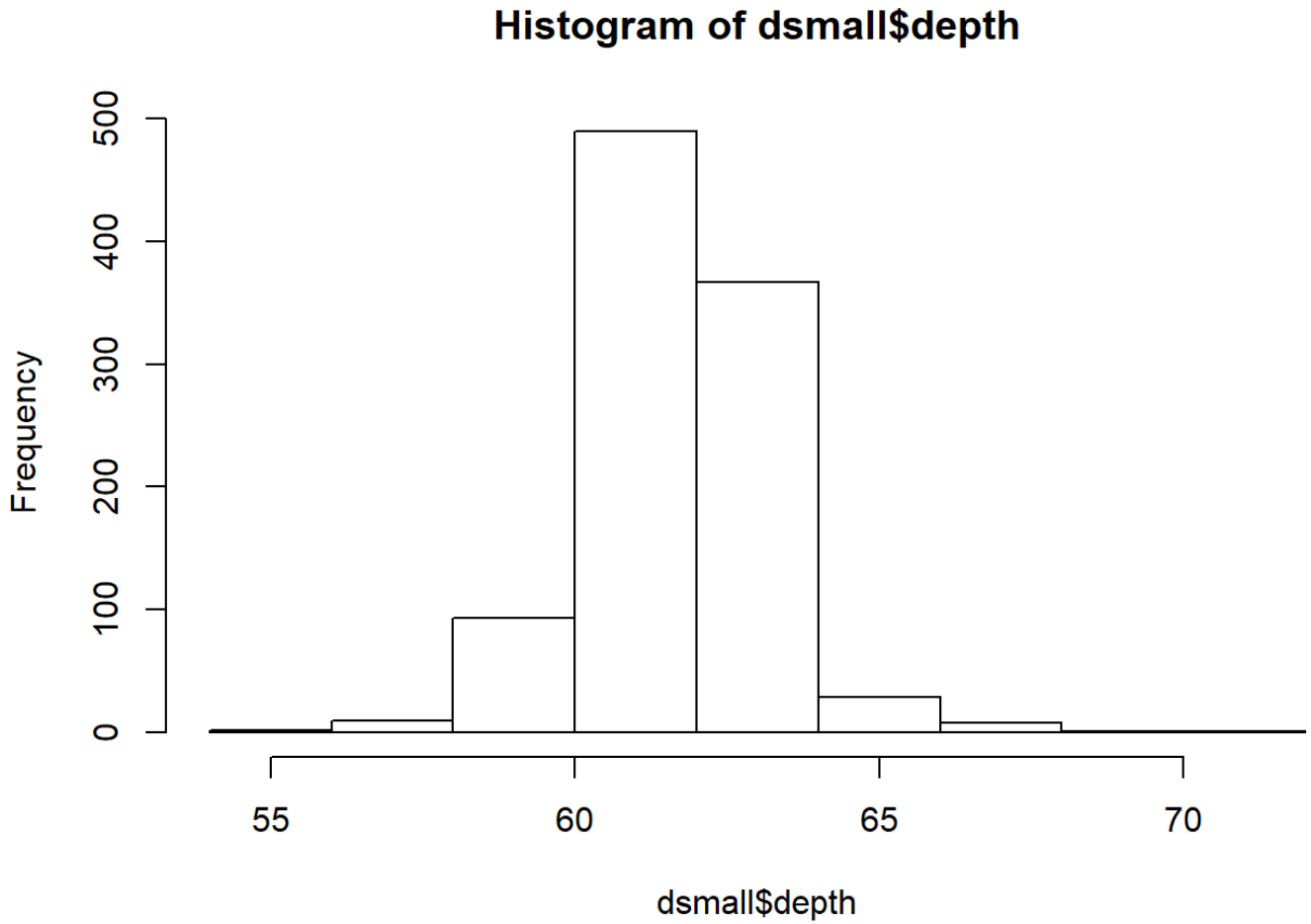
Obvious differences between box plots – see examples (1) and (2), (1) and (3), or (2) and (4). Any obvious difference between box plots for comparative groups is worthy of further investigation in the *Items at a Glance* reports.

Box Plot Vs Histogram



Histograms

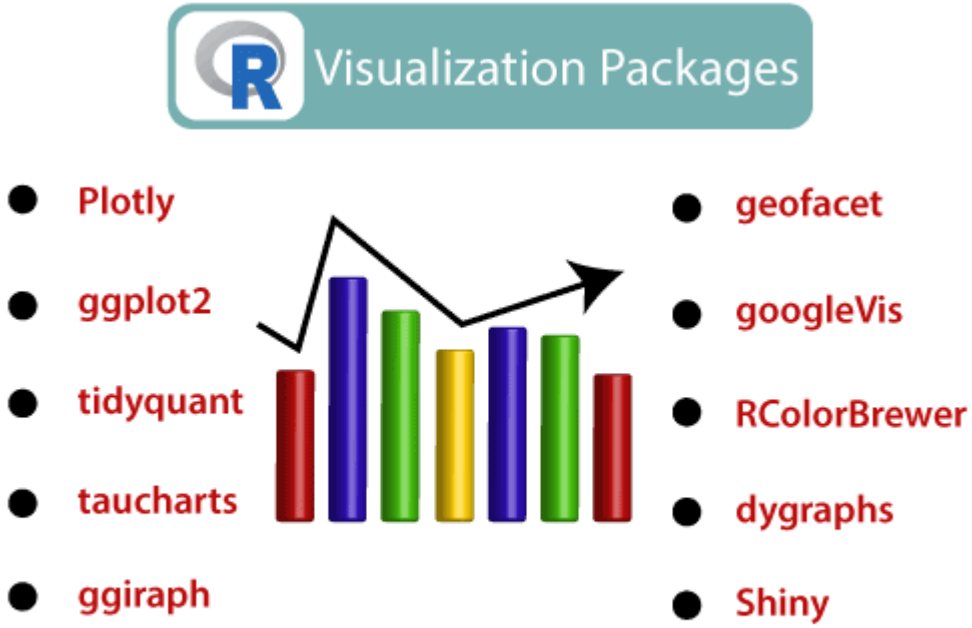
- This graph plots a distribution of numbers using a bar chart (with no spaces between the bars), representing the quantity of data that falls within a particular range. This visual makes it easy for an end user to identify outliers within a given dataset.



Data visualization with R

R Visualization Packages

R provides a series of packages for data visualization. These packages are as follows:



ggplot2

The basics of the grammar of graphics(ggplot2

There are some key elements of a statistical graphic. These elements are the basics of the grammar of graphics. Let's discuss each of the elements one by one to gain the basic knowledge of graphics.

